

EcoSmart AI

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## 01 Problem

For Everyday individuals, Nigerian households, and small and medium-sized businesses with recyclable waste, as well as informal local recyclers, recycling companies, and recycling processing plants, the repeated problem is many people do not know what can be recycled, what their waste is worth, or how to easily find someone who will collect it and pay a fair price, so recyclable materials are often thrown away, burned, or left to clog drains and worsen flooding, while recyclers struggle to consistently find and collect clean, sorted materials because buyers and sellers of recyclable materials are not easily connected, and information about what those materials are worth and where to take them is scattered.

## 02 Solution

EcoSmart AI and Innovation Ltd is building an AI-powered marketplace that identifies recyclable materials, estimates their value, and connects buyers and sellers directly for For Everyday individuals, Nigerian households, small and medium-sized businesses, and local recyclers, recycling companies, and recycling processing plants to achieve to achieve a simple, reliable way to sell, buy, and recover recyclable materials at fair value, reducing waste burned or clogging drains and creating income for both sides.

## 03 UVP

We help individuals, households, small and medium-sized businesses, and recyclers, recycling companies, and recycling processing plants in Nigeria do easier trading of recyclable materials better by identifying materials, estimating their value, and connecting buyers and sellers directly in one platform, unlike single-buyer platforms such as Chanja Datti, Wecyclers, TrashCoin, and Ecobarter that limit buyer choice and pricing transparency.

## 04 Unfair Advantage

A competitor starting today would need Months of field work to build genuine trust with recyclers and waste generators, plus enough real transaction data to train an accurate AI valuation model. because Trust in this industry is hard-won, as shown by real cases like Muhammad losing 9 million naira to a fraudulent deal, and no existing dataset for Nigerian recyclable material identification and pricing currently exists to shortcut this..

## 05 Customer Segments

Primary ICP: Waste buyers and collection aggregators who buys recyclable materials from scavengers and sometimes directly from schools, companies, households, and individuals, then sorts and sells them to recycling companies. in Gwagwalada, Abuja, Nigeria, behavior: About 70% of the recyclable materials he buys come from scavengers, who are his main source. He buys from them and resells the materials to recycling companies, but said he does not make as much profit because he is buying through another person. He sometimes gets calls from schools, companies, households, and individuals who want to sell their recyclable materials directly to him. This happens less often, but he said he can make about 60% more profit when he buys directly from the source. The materials he deals with include PET plastic bottles, aluminium cans, malt cans, brass, cardboard, and other metals. He sometimes has to borrow money or take a loan to pay suppliers upfront before he can resell the materials., JTBD: Find recyclable materials reliably, buy more directly from the source when possible, and sell to trustworthy recycling companies at a fair price

without being cheated or losing profit when prices change..

## 06 Key Metrics

Key Metric 1: GMV (Gross Marketplace Value) - Sum of all transactions via in-app wallet - Target: N500,000–N1,000,000 by Month 6

Key Metric 2: Commission Revenue -  $\text{GMV} \times 3\text{--}4\%$  - Target: N15,000–N40,000 by Month 6

Key Metric 3: Direct Sourcing Rate -  $\frac{\text{Direct-sourced materials}}{\text{total materials sourced}}$  - Target: Above 30% baseline

## 07 Channels

Primary channel: For waste aggregators and recyclers, we reach new customers through referrals, existing industry networks, and direct visits to physical collection and recycling sites for in-person sensitisation and onboarding. For individuals, households, and SMEs, we acquire users through Facebook and WhatsApp communities, TikTok and Instagram, and referrals, and promotion through Instagram, TikTok, and Facebook.. Trust tactic: Show a verified badge, number of completed transactions, and buyer/seller ratings on every profile.. CAC assumption: 2,000 to 5,000 naira per user onboarded.

## 08 Cost Structure

COGS: Naira 35,000

Operation Cost: Naira 650,000

Customer Acquisition Cost: Naira 1,175,000

Other Costs: Naira 7,200,000

Total Cost: Naira 9,060,000

## 09 Revenue Streams

Model 1: Commission = Naira 26,250